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MOTIVATION



- To develop systems that monitor urban noise pollution and classify the major contribution factor



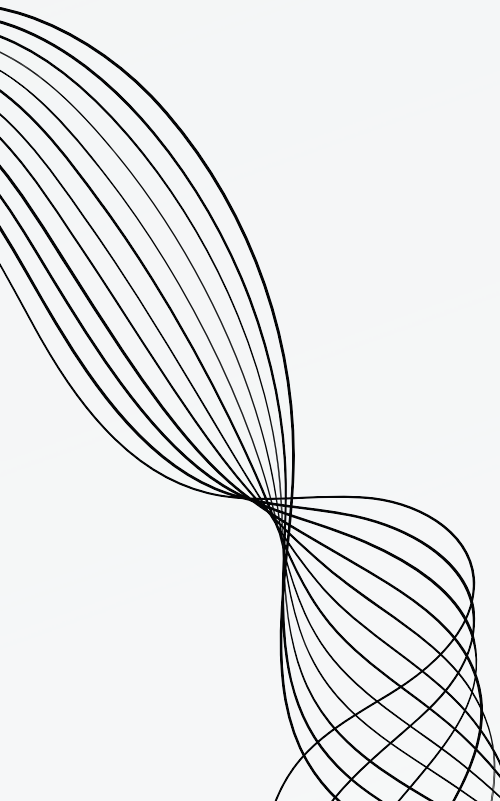
- Improve public safety and surveillance (classifying gunshots, explosions or cries)
- Help with environmental monitoring and wildlife conservations.





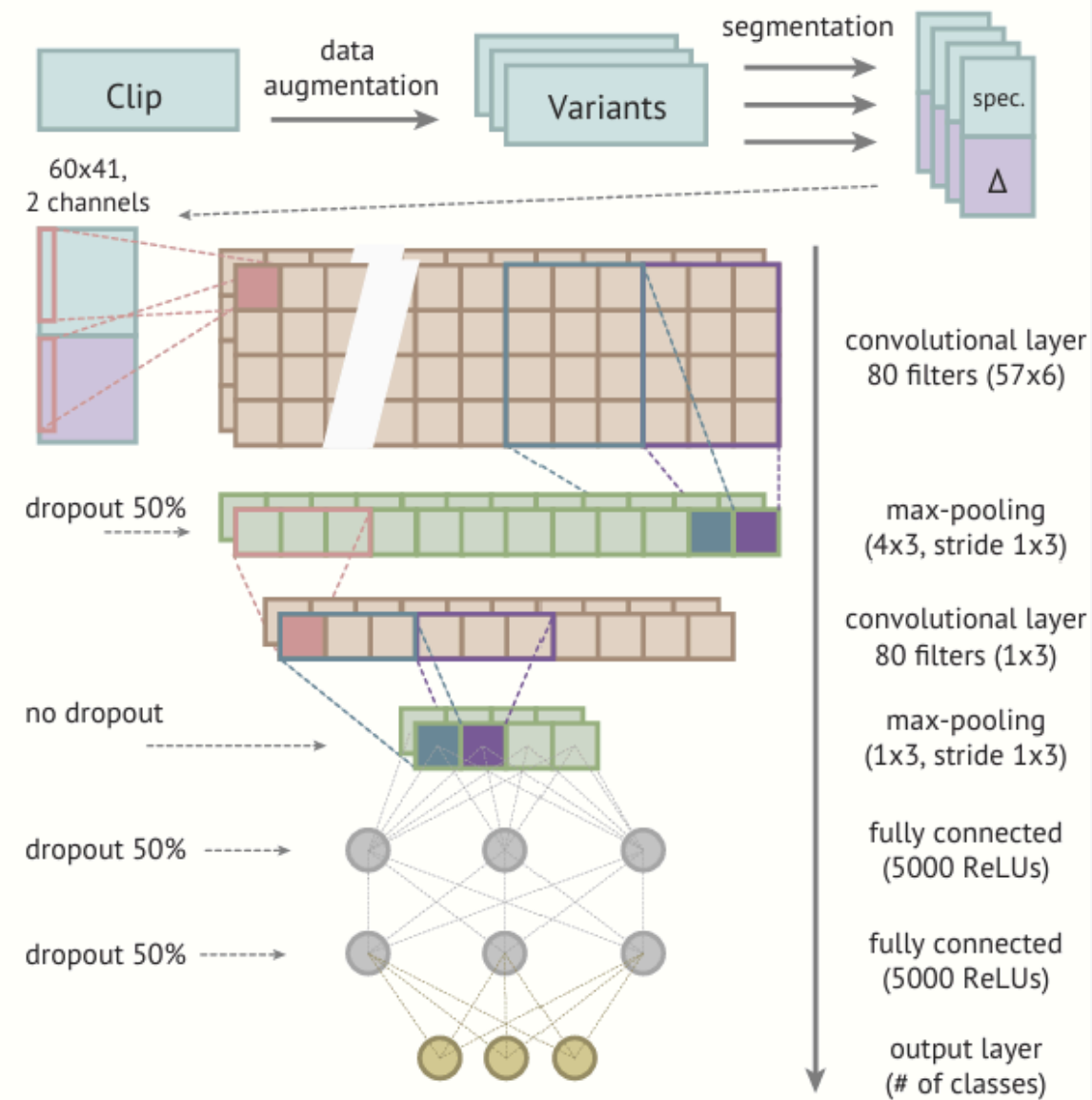
PROBLEM STATEMENT

To explore techniques for developing a robust and accurate system that can efficiently classify a relatively small number of classes using extremely limited datasets without pre-training.



LITERATURE REVIEW

Piczak-ConvNet



This CNN Baseline was proposed by the datasets author Karol J. Piczak, who focused on the potential of CNNs in classifying short audio clips of environmental sounds.

Model:

- 2 convolution layers with max pooling
- 2 fully connected layers

Data Augmentation:

- Applies random time delays, class-dependent time stretching and pitch shifting to original recordings before the model starts training

Data pre-processing:

- Log scaled mel spectrograms (using *librosa*), split into 41 or 101 frames at random with overlapping segments 50% and 90% respectively

Reported baseline accuracy of ~75%, evaluated using 5-fold cross-validation

Potential to build on:

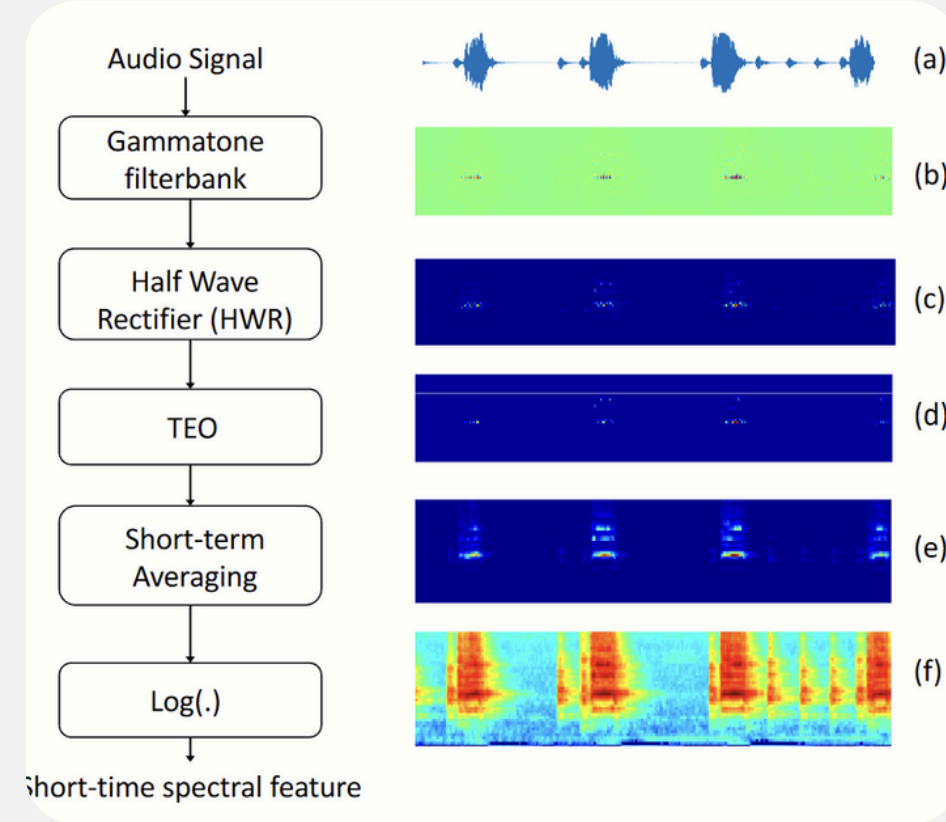
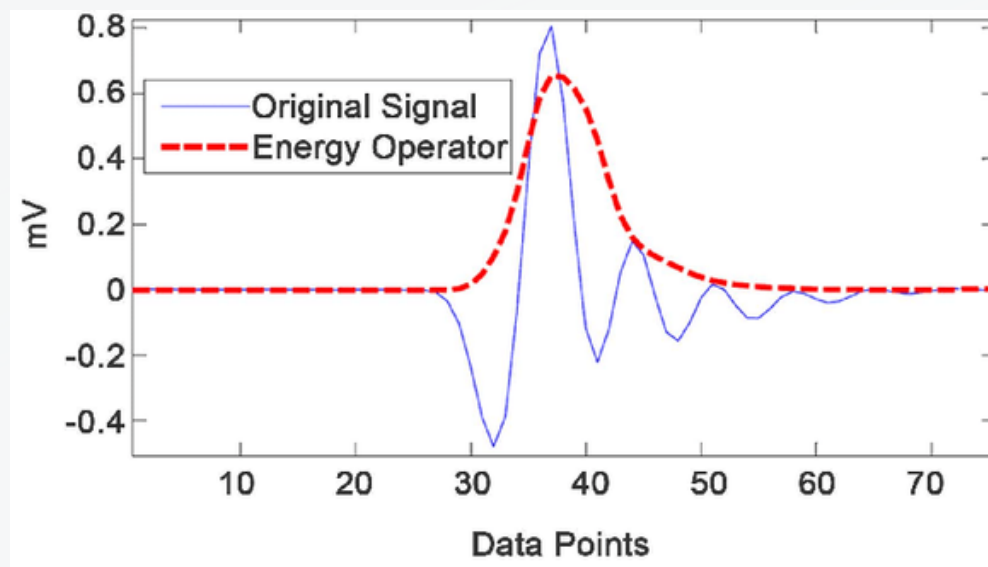
- applying augmentations on the fly for each batch
- no normalisation on the data
- better representation of audio features

LITERATURE REVIEW

TEO

Novel TEO-based gammatone features for ESC by Dharmesh M. Agrawal, Hardik B. Sailor, Meet H. Soni, and Hemant A. Patil.

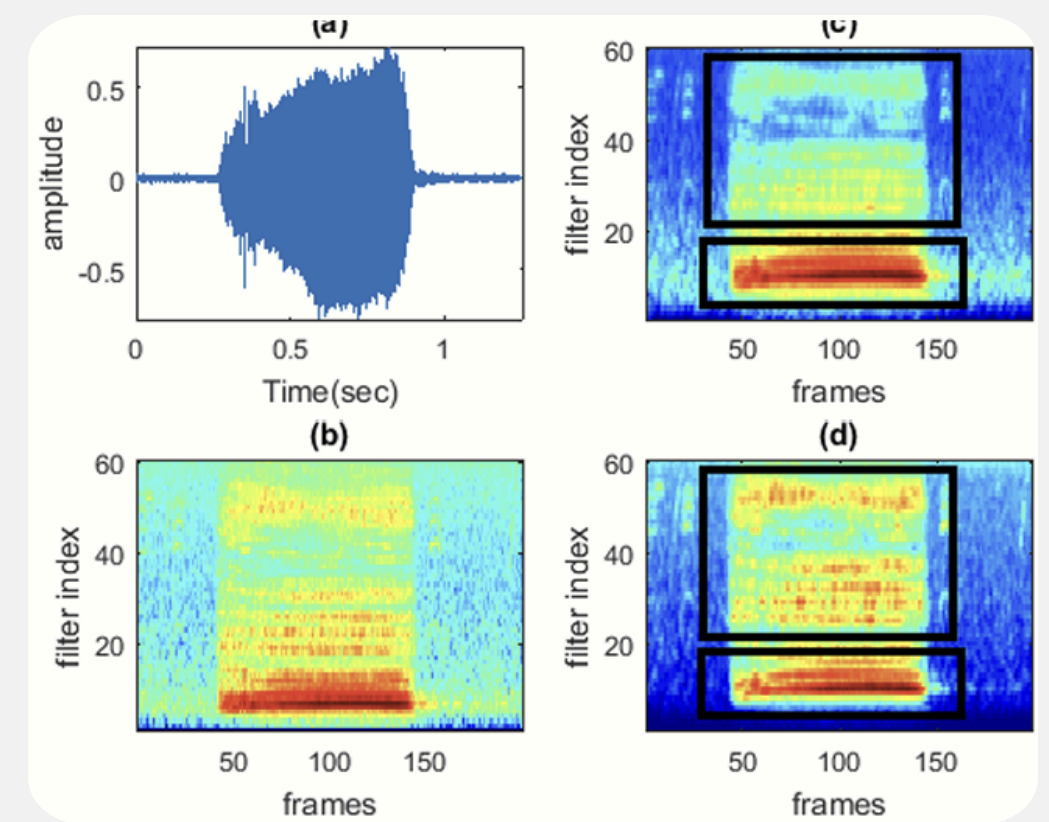
This paper claims only to use a novel data pre-processing approach and no augmentation despite achieving 81% accuracy using a slightly modified version of the CNN baseline proposed by Piczak.



Gammatone Filter takes the raw signal process and transforms it into a response which mimics the auditory processing of a human ear.

$$g(t) = at^{n-1}e^{-2\pi bt} \cos(2\pi ft + \phi),$$

The half wave rectifier further processes the audio signal by only taking the positive half of the audio signal similar to DC current



The resultant audio signal is then processed by applying Teager Energy Operator (TEO). This technique emphasises the transient features of the audio signal. This is done with a special measurement of energy that emphasises the variations in frequency and amplitude of the audio signal.

$$\Psi_d\{x[n]\} = x^2[n] - x[n-1]x[n+1] \approx a^2[n]\omega^2[n]$$

Log magnitude and short-term averaging are utilized to adjust the audio signal to have a correct scale and it has proven to show greater accuracy for training the CNN.

LITERATURE REVIEW

ACD-NET

Aimed at introducing a pipeline for sound classification using deep learning in extremely resource constrained (low energy, low memory and lower CPU speed) micro controllers.

Pipeline:

CNN architecture
(ACDNet ~ 96.65% accuracy on ESC-10)

->

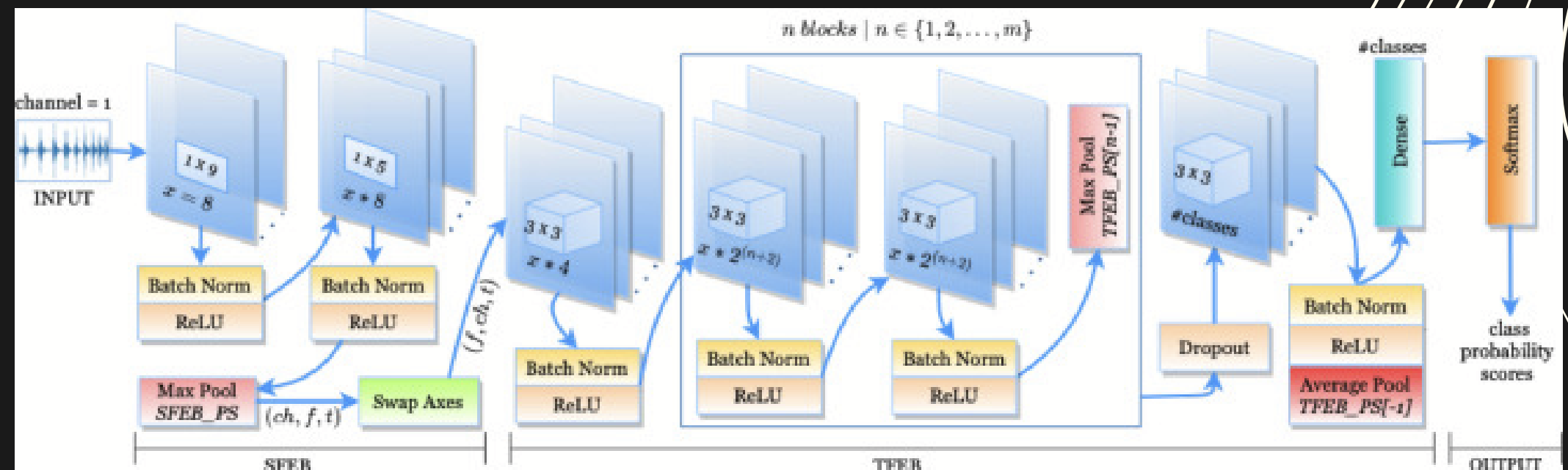
Hybrid compression technique
(sparsify weight matrices then apply)
then apply structured compression

->

Micro-ACDNet
(97.28% fewer flops and 97.22% smaller in size)
96.25% on ESC-50

ACDNet Architecture:

- Convolution layers -> batch normalisation -> RELU
- Maxpooling (stride equal to pool size to avoid overlapping)
- Two feature extraction blocks:
 - Spectral feature extraction (SFEB)
 - low level audio features
 - two 1d strided convolutions
 - Temporal feature extraction (TFEB)
 - high level features
 - convolves over both frequency and time



Differences with other papers:

- Significantly larger initial proposed model at 101M parameters making it difficult to train
- Compression process prunes off above 90% of "useless" nodes, so for using it on a smaller well trained model either will not maintain accuracy or not reduce size by significant amount

DATASETS

- The environmental sound classification dataset **ESC-10 (subset ESC-50)**

Overview:

- 400 short (5-seconds) environmental recordings
- Grouped into 5 major classes (animals, nature, human, domestic, urban noises) with 10 semantic classes
- Prearranged into 5 folds for reliable cross-validation
- Balanced dataset equal number of instances in all classes

Reference benchmarks:

- Human benchmark ~96% on untrained participants

As ESC's are largely unstructured and highly diverse, possible features for a model to pick up on are:

1. Frequency & Amplitude

- The model will have to differentiate between different strengths and frequencies of the same sound as events such as snoring, animal noises and water drops can differ immensely based on how they are recorded.
- With 40 examples per class it is likely variations will match with other potential sounds which is difficult to differentiate (human accuracy fell on similar softer sounds)

2. Silence

- Some unstructured sounds like sea waves and snoring have periods of minimal audio signal (silence) - the model must recognize
- Others will have large silent periods before and after the noise such as can opening and a door knock (zero padding)

As the variety of features is still large and diverse on ESC-10 is a acceptable choice for a dataset to train/ test the robustness and accuracy of the model while being able to overcome the large computational cost of ESC-50. Where on Average it takes approximately 5x longer per epoch

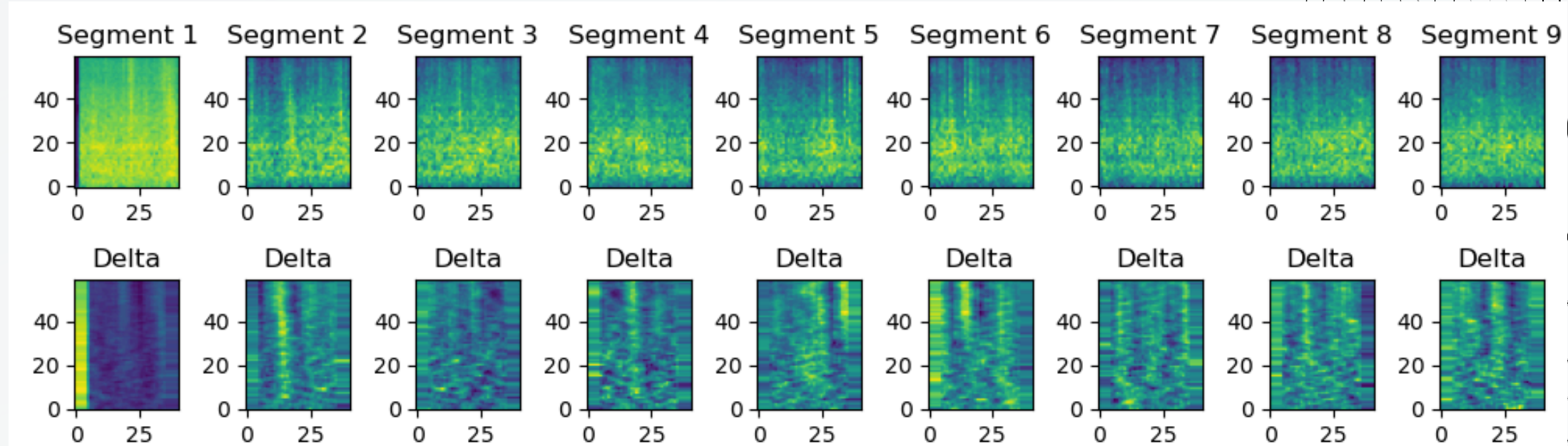
DATA ANALYSIS

Log-Mel

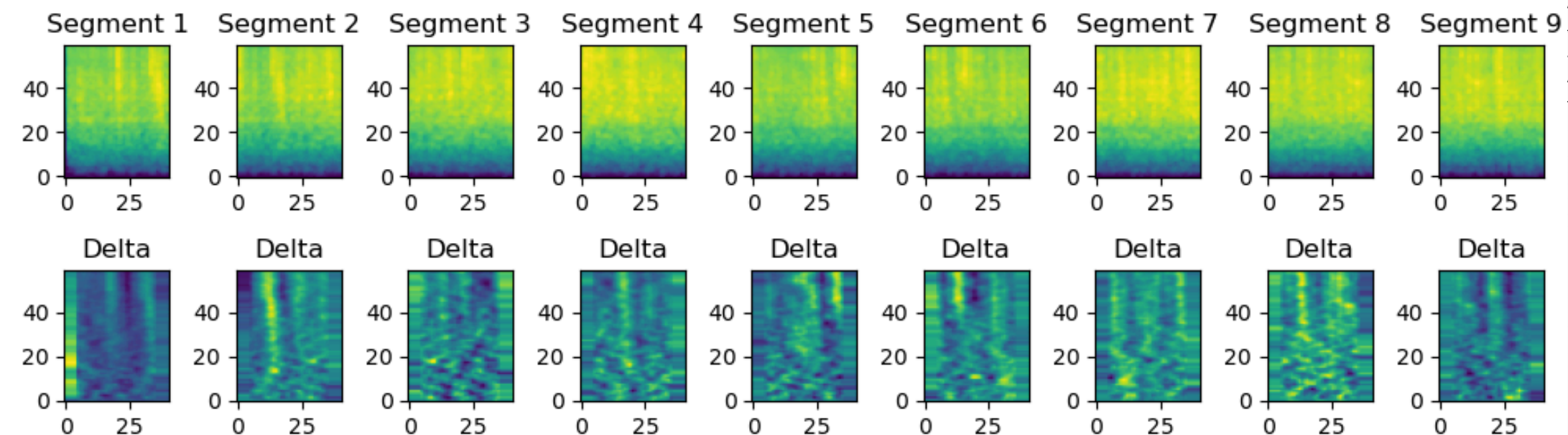
- Possible Data Representations: Mel, Log-Mel, Gammatone-Gram, TEO-GTSC (pure GTSC couldn't be replicated due to little info given)

Advantages of TEO-GTSC over Gammatone-GRAM and log mel:

- More features are visible compared to mel spectrogram output
- high differentiability between features, more information present to learn differences
- Higher resolution of features, especially in the low-frequency regions: see delta for easier distinguishing with the human eye



Pure TEO-GTSC



METHODOLOGY

To keep the training and inference cost down we will have to keep the DNN shallow. So to compensate, we have

Fold Splitting
Split raw ESC-10 dataset
into 5 folds for cross-
validation

1

Data Pre-Processing
Pre-process the raw
waveforms using different
techniques

2

(Ensemble Evaluation
Supported)
Probability voting across
segments improves
generalisability and predictions
of several classifiers.

Model Training &
Evaluation

Test different combinations of
architectures and datasets
along with improvements made

3

To test models and baselines in papers everything was to be made from scratch to use our custom framework where we can plug in different datasets & networks that we replicate/invent using a uniform interface.

This helped us to achieve better consistency when testing claimed performance and evaluating our changes to the model compared to benchmarks and other previously implemented works in the literature.

Data augmentation was not performed:

- Significantly less training time
- No significant improvement for limited datasets, only overhead
- This is backed by research (due to lack of time & funding we'll take their word for it)

PREPROCESSING

Steps of signal processing to increase feature dimensions
(and add linear separations)

Filtering

We apply Gammatone Filterbank to extract the peak frequency deltas, positively rectified.

1

TEO

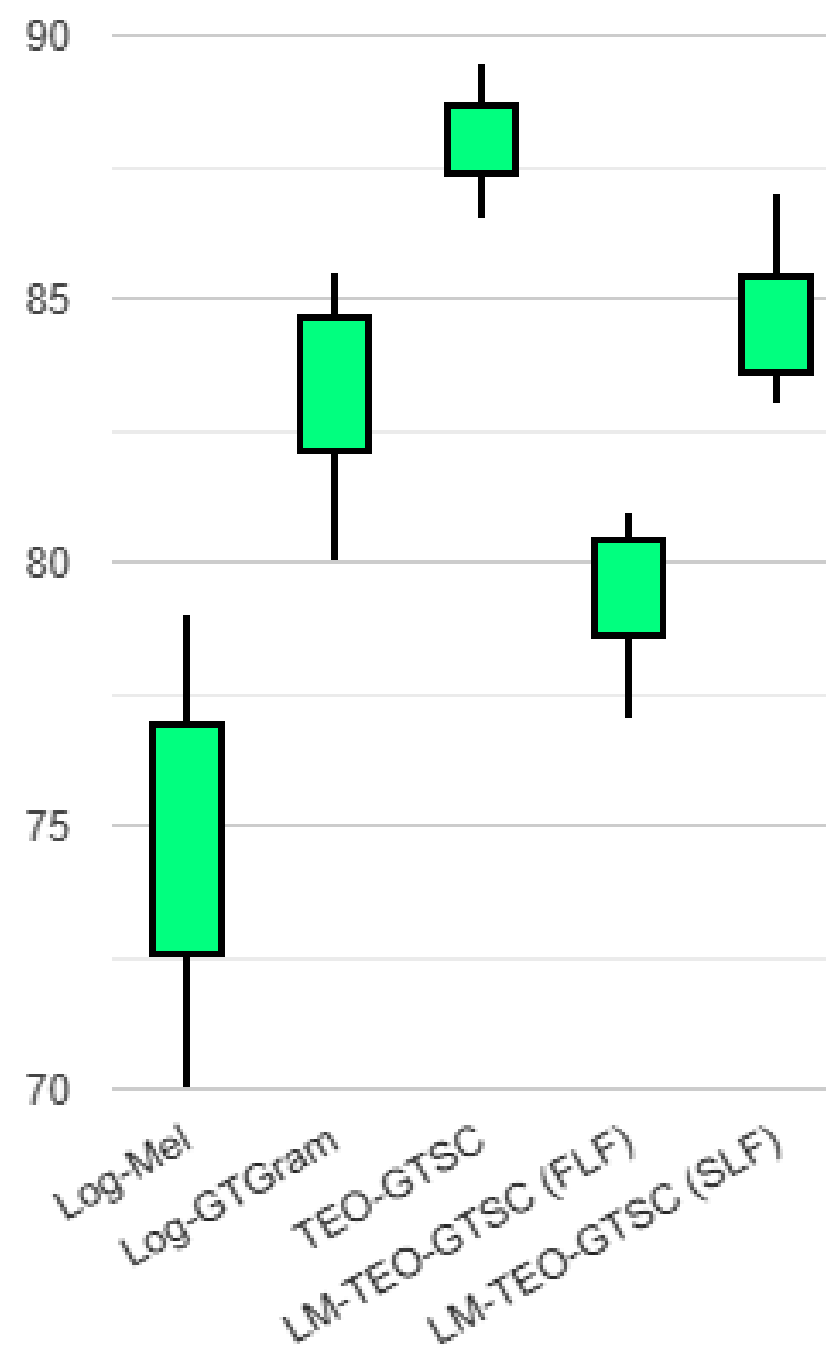
Apply Teager Energy Operator to

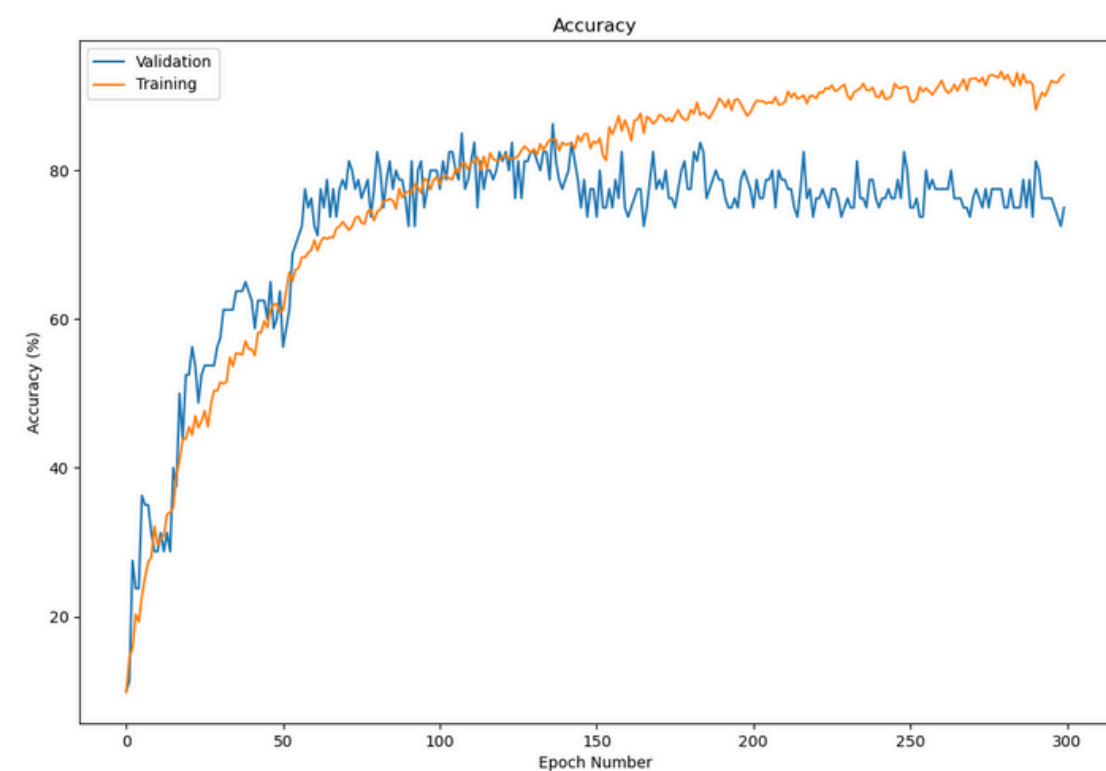
2

Apply log to output tensors to create a log scale as frequencies range from 0 to 22.5k Khz

3

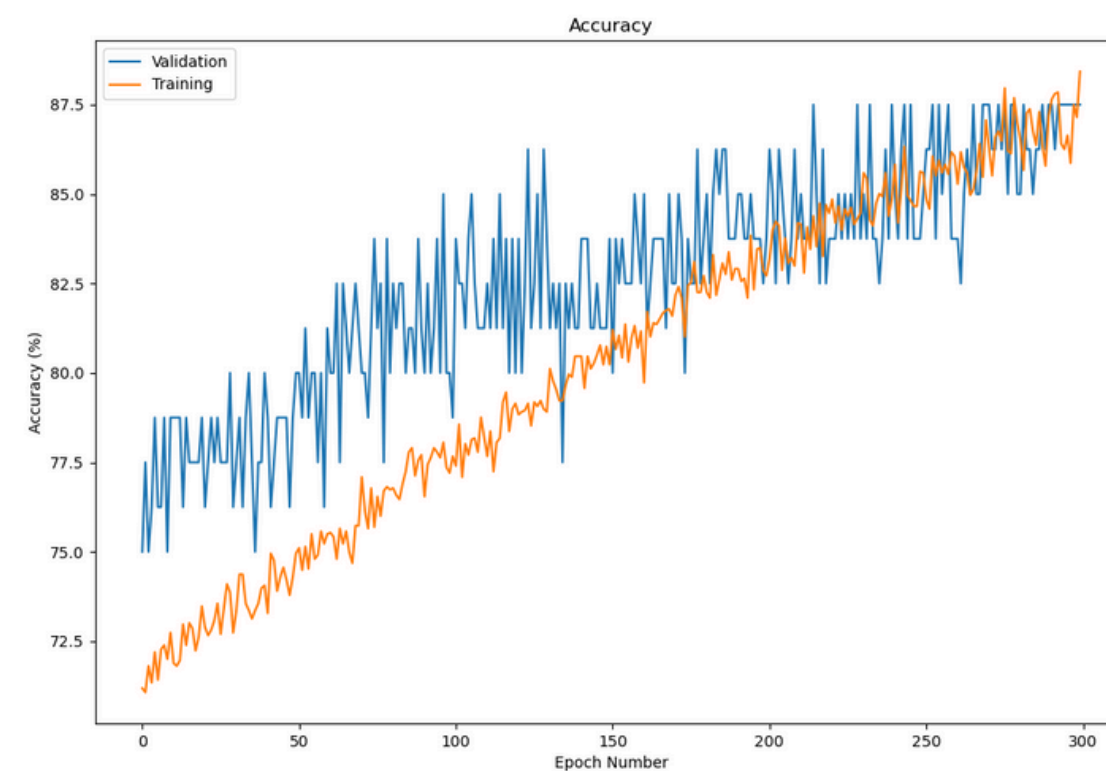
Preprocessing Accuracies on Piczak-CNN



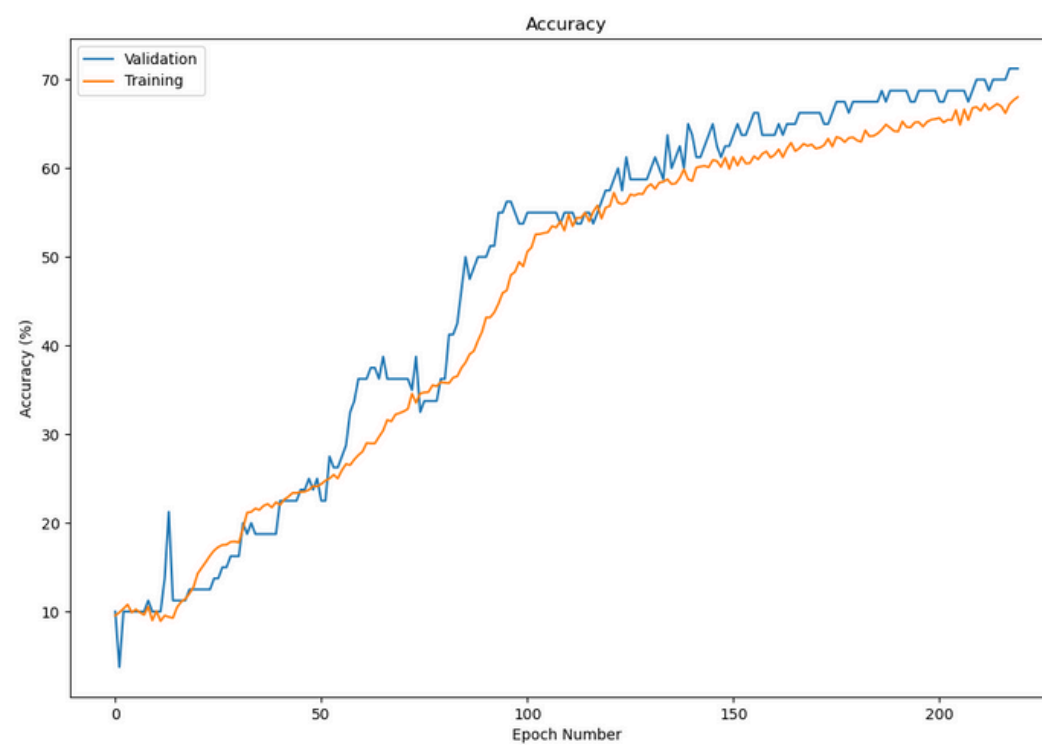


Log-Mel

Plot Accuracies



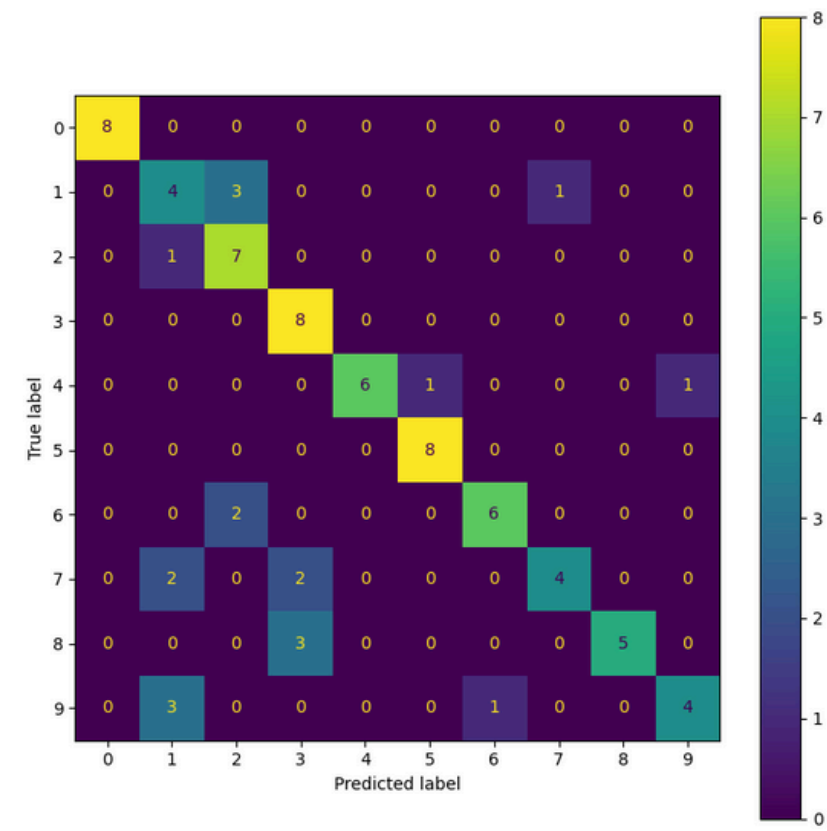
TEO-300-600



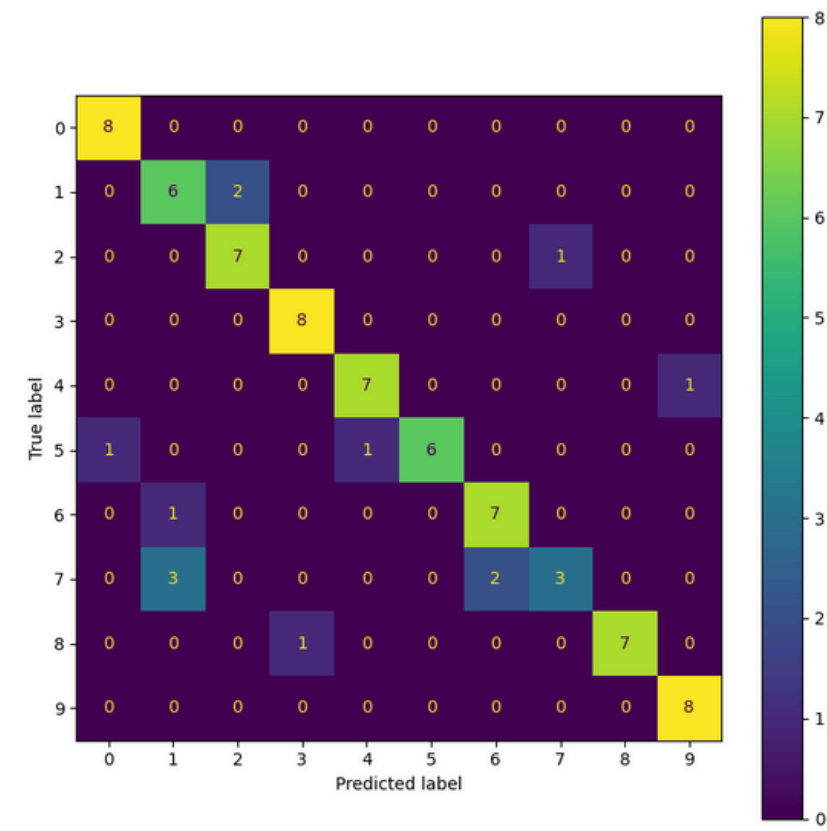
TEO-300

Confusion Matrix

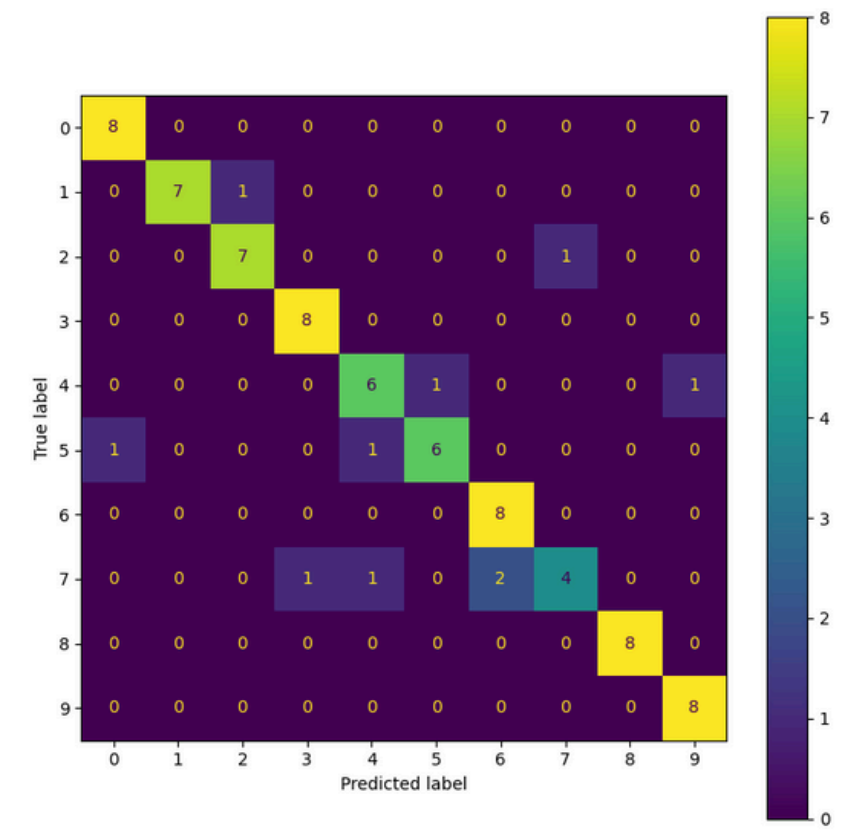
Log-Mel



Gammatone Filterbank



TEO-GTSC



RESULT

A circular progress indicator with a yellow center and a grey outer ring, showing 90% completion.

90%

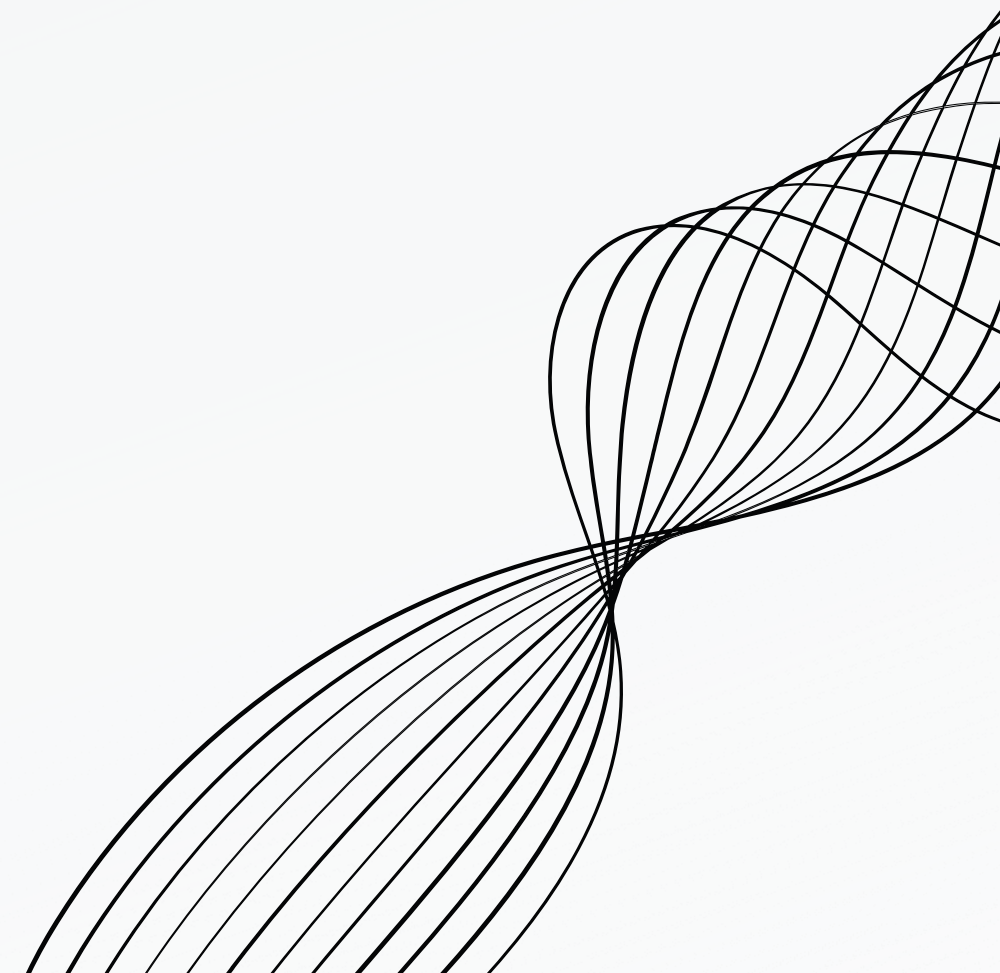
ACCURACY ~
90% FOR (TEO-
GTSC)



TIME
5S PER EPOCH
TO TRAIN ON
3060 ~ 50
MINUTES FOR
600 EPOCHS



SIZE
~26,734,090
LEARNABLE
PARAMETERS



CONCLUSION

**ANY
QUESTIONS?**

